

Dr. NKODIA Hardy Medry Dieu-Veil

Geologist — Tectonics, geodynamics and structural geology

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Profile

Lecturer–researcher in structural geology and tectonics: palaeostress analysis, fault reactivation and intraplate seismic hazard in Central Africa. *Interests:* fault/fracture reservoirs (Inkisi, Pool); Congo Basin geomorphology; mineralisations and aquifers; seismic hazard of the Central African rifted margin.

Education

2019 – 2024	PhD in Geosciences — Marien Ngouabi University (Congo) & Royal Museum for Central Africa (Belgium). Highest distinction with the jury's congratulations.
2015 – 2017	MSc in Geosciences — Marien Ngouabi University. Distinction; top of class.
2012 – 2015	BSc in Geosciences — Marien Ngouabi University. 2nd of class.
2011	Baccalauréat (series D) — Lycée Chaminade, Brazzaville.

Academic appointments

2025 – present	Assistant (tenure-track, toward Assistant Professor) — Marien Ngouabi University, FST, Dept. of Geology (Congo). Teaching and research in structural geology; MSc supervision.
2025 – 2026	Postdoctoral researcher — Pukyong National University, Busan (South Korea), on research leave from Marien Ngouabi University. Contemporary stress field and seismogenic structures of the Korean Peninsula.
Aug – Sep 2026	Research visit — RMCA, Tervuren (Belgium). West Congo Belt and karst.
Feb–Mar 2025	Postdoctoral researcher (visit) — Royal Museum for Central Africa (RMCA), Tervuren (Belgium).
2018 – 2024	Lecturer–Researcher — Marien Ngouabi University (Congo). Structural Geology course (L2); co-organised an international conference.
2019 – 2021	Research assistant — RMCA, Tervuren (Belgium). Managed a €15,000 budget over 3 years; international collaborations.
2019 – 2021	Instructor — African School of Development (EAD), Brazzaville. Structural geology course.

Grants & honours

- **AMORCE-GEO (2025–2027)** — project funded by ARES (Belgium); role: co-organiser and trainer.
- PhD awarded with the jury's congratulations (2024); top of MSc class (2017).

Publications — selected (peer-reviewed)

Full list: 12 peer-reviewed articles (see ORCID / Google Scholar).

1. **Nkodia, H. M. D.-V.**, Hategekimana, F., Naik, S. P., Cheon, Y., Peace, A. L., Bae, S.-Y., Kandregula, R. S., & Kim, Y.-S. (2026). Vertical kinematic partitioning in intraplate fault systems: evidence from the SE Korean Peninsula. *Journal of Structural Geology*, **211**, 105778. doi:10.1016/j.jsg.2026.105778
2. **Nkodia, H. M. D.-V.**, Park, K., Naik, S. P., Peace, A. L., & Kim, Y.-S. (2026). Assessing the contemporary stress field and seismogenic structures of the Korean peninsula using focal-mechanism inversion and slip-tendency analysis. *Tectonophysics*, **934**, 231261. doi:10.1016/j.tecto.2026.231261
3. Colet, M., Kolawole, F., Ajala, R., Delvaux, D., & **Nkodia, H. M. D.-V.** (2025). Active crustal deformation across a nucleating extensional microplate, D. R. Congo, East Africa. *Tectonics*, **44**, e2025TC008815. doi:10.1029/2025TC008815
4. **Nkodia, H. M. D.-V.**, Boudzoumou, F., Miyouna, T., *et al.* (2024). Brittle faulting and tectonic stress history on the western margin of the Congo Basin between Kinshasa and Brazzaville. *Tectonophysics*, **877**, 230282. doi:10.1016/j.tecto.2024.230282
5. Mavoungou, Y. B., **Nkodia, H. M. D.-V.**, Watha-Ndoudy, N., & Bolarinwa, A. T. (2024). Lineament extraction and paleostress analysis in the Bikélélé iron deposit (Chaillu Massif, Congo). *Modeling Earth Systems and Environment*, **10**(2), 1993–2009. doi:10.1007/s40808-023-01883-3
6. **Nkodia, H. M. D.-V.**, Miyouna, T., Kolawole, F., *et al.* (2022). Seismogenic fault reactivation in western Central Africa: insights from regional stress analysis. *Geochemistry, Geophysics, Geosystems*, **23**(11), e2022GC010377. doi:10.1029/2022GC010377
7. **Nkodia, H. M. D.-V.**, Miyouna, T., Delvaux, D., & Boudzoumou, F. (2020). Flower structures in sandstones of the Palaeozoic Inkisi Group (Brazzaville, Congo). *South African Journal of Geology*, **123**(4), 531–550. doi:10.25131/sajg.123.0038
8. Miyouna, T., **Nkodia, H. M. D.-V.**, Essouli, O. F., *et al.* (2018). Strike-slip deformation in the Inkisi Formation, Brazzaville, Republic of Congo. *Cogent Geoscience*, **4**(1), 1542762. doi:10.1080/23312041.2018.1542762

Invited talk

- **6 July 2026** — “Ocean–continent tectonic stresses and earthquake hazards of the Central African rifted margin”. Online seminar *Earthquake Hazards of Rifted Margins*, Columbia University / Lamont-Doherty Earth Observatory. Recording.

Skills, languages & service

- Tectonic/structural analysis and palaeostress (slip tendency); remote sensing & GIS (ALOS-PALSAR DEM, satellite/drone imagery); petrology and field geology; Python (publication figures), Adobe Illustrator.
- *Certifications*: Project Management (Google, 2024); scientific writing/publishing (École Polytechnique Paris, 2023); radar remote sensing (IGNFI/AFD, 2019).
- *Languages*: French (native), English (professional). *Service*: Founder & President of Kongo Science; Founder of the LGRN (Marien Ngouabi University).

References

- **Prof. F. Boudzoumou** (MNU).
- **Dr. D. Delvaux** — Editor-in-Chief, *J. African Earth Sci.*; RMCA, Tervuren.
- *Referees' contact details available on request.*